

Basic Arthroscopic Principles

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The first arthroscope was developed in 1920 with an optical cannula diameter of 7.3 mm by Dr. Kenji Takagi from Tokyo. His idea to look inside a closed knee with an instrument, a cystoscope, in 1918 came from his interest in the early diagnosis of tuberculous knees in Japan, which was a problematic disease at the time. Dr. Takagi is credited with being the first innovator and developer of arthroscopy. His successor, Dr. Masaki Watanabe, continued advancing Dr. Takagi's arthroscope design, which contributed largely to the popularization of arthroscopy. He also developed the concept of triangulation. Prior to Dr. Takagi, Dr. Nordentoft, a Danish surgeon, in 1912 published a manuscript at a meeting in Berlin, where he adapted the use of an endoscope to the knee joint and is credited as being the first to apply endoscopy to the knee, as well as the first to use the term "arthroscopy."¹

Arthroscopy has revolutionized the treatment of intraarticular pathology. Arthroscopy has become the standard of care for many orthopedic problems and is currently the most common orthopedic procedure performed in the United States, with 1 million cases reported annually.² Arthroscopy in experienced hands has low morbidity and high diagnostic yield. The benefits of arthroscopic surgery include reduced morbidity, less post-operative inflammation, smaller incisions, improved diagnostic accuracy, lower complication rates, reduced hospital stay, and reduced cost.³ The decision to use arthroscopy must involve consideration of the potential benefits of this method weighed against its limitations.

ARTHROSCOPIC APPLICATIONS

Advancements in arthroscopic technology have expanded its use in nearly every joint in the body, including those in the elbow, wrist, ankle, foot, and spine. Although results in some areas are superior to those of open surgery, such superiority has not been demonstrated for all arthroscopic procedures. The primary applications of arthroscopy are the diagnosis and treatment of intraarticular pathology. Arthroscopy is commonly used in the knee and shoulder for irrigation and débridement, synovectomy, removal of a foreign or loose body, débridement or repair of osteochondral lesions, and ligament reconstruction. Arthroscopy has also been applied to the hip for treatment of femoroacetabular impingement with repair or reconstruction of the labrum, cartilage, osseous deformity, and capsule.⁴ Absolute contraindications to arthroscopy are skin infection over the operative site or at a remote site, with risk of seeding. Relative contraindications are

ankylosis of the joint and major capsular disruptions that risk excessive extravasation of fluid and make joint distension difficult.

EQUIPMENT

Arthroscopy is performed in a standard surgical suite in a hospital or ambulatory surgical center. Arthroscopic equipment has become more streamlined, occupying less space in the operating room. Some operating rooms are now specifically made for use with arthroscopy and are fully equipped with mobile monitors and an arthroscopic tower attached to the ceiling along with the standard operating room lights (Fig. 8.1). The operating room must be large enough to accommodate the required equipment and should be staffed by operating room personnel experienced with arthroscopy.

Arthroscopic cables and equipment are sterilized mostly by steam sterilizing autoclaves. The word autoclave comes from Greek roots auto (self) and clave (key), which together means self-locking. Autoclaves have an automatic locking ability that allow temperatures to exceed 270°F and pressures of 15 psi, which are the recommended settings for steam sterilization, creating an environment that no organism can survive in. The arthroscopic sterilization process begins with a standard wash of all the instruments, which then get wrapped with a blue sterilization wrap and sterilization indicator tape. Along with the sterilization indicator tape, a steam chemical indicator strip is attached to each individual tray and a Bowie-Dick Plus Test Pack is placed alongside the entire load to be autoclaved to ensure complete sterilization. These indicators turn black once fully autoclaved. A biologic indicator is also placed with the autoclave set, then inserted into a steam incubator and compared against a control biologic indicator for 1 hour to ensure all organisms have been eliminated. A plus sign for the control indicator and minus sign for the autoclaved indicator are digitally displayed to ensure complete sterilization. These sterilization tests are all documented and kept on record by the OR staff. For temperature-sensitive instruments such as smaller cameras and plastics, alternate methods of sterilization are used such as gas sterilization with ethylene oxide, low-temperature sterilization using peracetic acid (Steris, Mentor, OH), and cold disinfection using activated glutaraldehyde (Civco, Kalona, IA).⁵

Arthroscope

Advances in technology have resulted in improved arthroscope design and function. The arthroscopes currently used have

Abstract

Arthroscopy has revolutionized the treatment of intraarticular pathology. Arthroscopy has become the standard of care for many orthopedic problems and is currently the most common orthopedic procedure performed in the United States, with 1 million cases reported annually. Arthroscopy in experienced hands has low morbidity and high diagnostic yield. The benefits of arthroscopic surgery include reduced morbidity, less postoperative inflammation, smaller incisions, improved diagnostic accuracy, lower complication rates, reduced hospital stay, and reduced cost. The decision to use arthroscopy must involve consideration of the potential benefits of this method weighed against its limitations. In this chapter we will discuss arthroscopic applications and the instruments and equipment used to perform arthroscopy. We also reviewed the latest research on arthroscopic techniques and complications associated with arthroscopic procedures.

Keywords

arthroscopy
arthroscopic principles
fluid pump
arthroscopy instruments
arthroscopic complications

improved optics and field of view with smaller diameter lenses, and they use fiberoptic and digital technology to improve visualization. Modern camera systems have the ability to use ultra HD 4K technology. Most arthroscopy systems allow video recording and the ability to take pictures. Newer technology integrating the camera system with tablet computers for arthroscopic video recordings, pictures, and editing will allow for higher-quality presentations and improved media for educational purposes.

The arthroscope is designed to fit inside a cannula, which is inserted into the joint first with use of a blunt trocar. Modern cannulas allow the flow of irrigation fluid into the joint. The camera is housed within the arthroscope and connected to the digital monitor for direct visualization of the joint. The light source, which comes from the light projector in the form of a

fiberoptic cable, is attached to the arthroscope to allow visualization and minimizes the heat generated at the arthroscope.

Arthroscopes have different optical characteristics, with several options in terms of lens diameter, field of view, and angle of inclination. The diameter of the lens determines the size of the arthroscope and the field of view and can vary from 1.7 to 7 mm. The field of view is represented by the total angle visualized by the lens of the arthroscope, with a larger lens diameter allowing a greater field of view. The angle of inclination is defined as the angle between the long axis of the arthroscope and a line perpendicular to the lens and can vary typically from 0 to 70 degrees. The advantage of using angled arthroscopes is that they provide better control and allow a wider field of view with rotation of the arthroscope, but a central blind spot directly in front of the arthroscope is created with arthroscopes that have a higher angle (70 to 90 degrees) (Fig. 8.2). The degree of magnification of an object viewed with an arthroscope is determined by its distance from the lens. This characteristic makes judging arthroscopic distances and sizes difficult, but the use of an object of known size for comparison (such as a probe) can be of assistance.

The most commonly used arthroscope is 4 mm in diameter and has an angle of inclination of 25 to 30 degrees. Smaller joints can be accessed using arthroscopes with a smaller diameter (1.9 or 2.5 mm). Arthroscopes with increased angles of inclination (70 or 90 degrees) can be used to view difficult areas such as the posterior corners of the knee.

The fiberoptic light source and power console for shavers and drills are typically housed on an arthroscopic cart. Such a cart can be positioned easily in the operating room so it is close to the patient but does not obstruct personnel. Arthroscopy using modern equipment requires a light source capable of producing 300 to 350 watts, for which tungsten, halogen, LED, and xenon arc sources have been adapted. Fiberoptic cable uses bundled glass fibers to transmit light from the source to the



Fig. 8.1 The arthroscopic cart with camera, monitor, light source, power for shaver and handpieces, and pump.

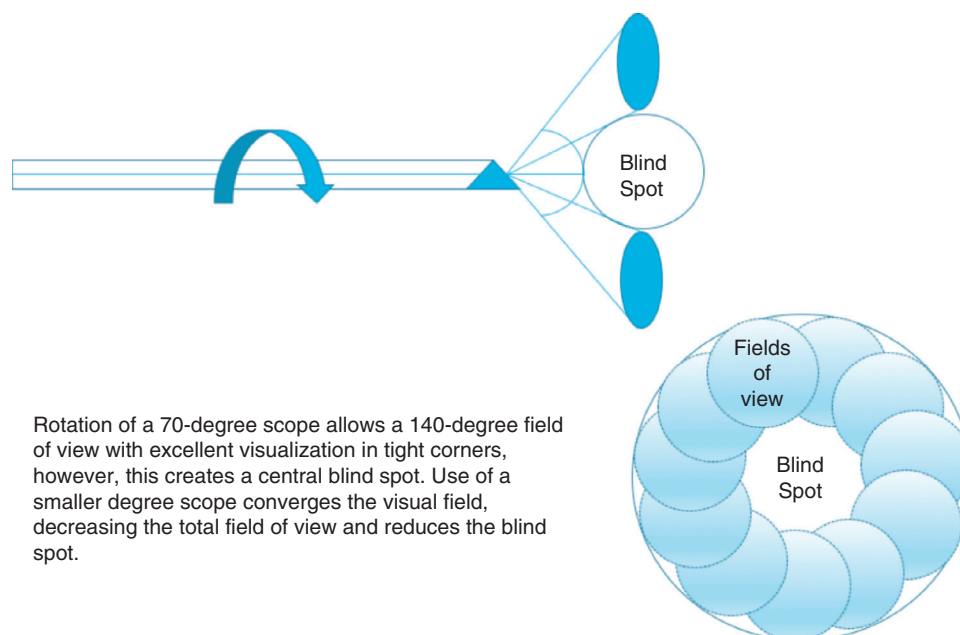


Fig. 8.2 The angulation of each scope determines the field of view as well as the potential blind spot.

arthroscope. The intensity and quality of light are diminished by the length of cable or damage to it, although modern fiber-optic cables are more durable.

Reduction of camera size, improved camera resolution, digitization of the video signal, and the integration of high-definition technology have all contributed to considerable improvement in the quality of visualization obtained using arthroscopy.

Fluid Pump

Irrigation fluid is vital for arthroscopy because it is responsible for joint distension, removal of debris, and improvement of visualization. Joint distension is supplied by the hydrostatic pressure of the irrigation fluid, which can be maintained using gravity, mechanical pumps, or manual control of outflow usually by the scrub tech. Every foot of elevation of the irrigation solution above the joint results in approximately 22 mm Hg of pressure. The use of a fluid pump allows more precise control of the rate of flow and hydrostatic pressure in the joint and can be used to minimize extravasation of fluids into the tissues.

Fluid management in arthroscopy consists of balancing the flow of fluids in and out of the joint. This balance is vital to maintain appropriate distension of the joint without causing excessive fluid extravasation into the tissues, and it plays a role in the management of bleeding, visualization, and debris clearance. Imbalances between inflow and outflow can cause turbulence of the irrigation fluid, reducing visualization and thus increasing operative time. This can occur when the shaver is used without compensatory changes in inflow/outflow or when the manual outflow of fluid is not controlled appropriately. Control of intraarticular pressure is critical, especially when working with the shaver in the posterior aspect of the knee. A situation where flow is decreased secondary to overzealous suction through the shaver can potentially deliver the posterior capsule into the shaver inadvertently. To prevent this problem, the inflow rate can be increased, the outflow through the suction or arthroscope can be decreased (or closed to prevent back flow), or outflow through the shaver can be decreased manually or by ensuring that the shaver window is occupied with tissue to be cut. Development of automated inflow and outflow mechanical irrigation pumps have shown promising results with consistent flow, greater degree of joint distension, and improved visualization, leading to reduced operative times.⁶ Modern irrigation systems also have functions that allow brief increases in pressure and flow to clear bleeding or debris, as well as functions to drain fluid such as at the end of the procedure.

A physiologic solution such as Ringer lactate or normal saline should be used for irrigation; the surgeon decides which solution to use. The addition of epinephrine to the irrigation fluid (1 mg/L) has been shown to shorten operating time and decrease bleeding, leading to improved visualization.^{7,8} Recently, it was shown that norepinephrine added to irrigation fluid reduces the incidence of hypotensive/bradycardic events significantly compared with epinephrine and is as efficacious in controlling intraoperative bleeding, thus maintaining clarity of the visual field when used in shoulder arthroscopy.⁹

The temperature and osmolality of the irrigation fluid used in shoulder arthroscopy has been a recent topic of discussion

in the literature. Three randomized controlled studies evaluated the effect of warmed irrigation fluid versus room temperature irrigation fluid. One study did not demonstrate a significant difference in reducing perioperative hypothermia, whereas the other two showed that hypothermia occurred more often in the room temperature irrigation fluid group.^{10,11} Kim et al. showed that age and the amount of irrigation used correlated with core body temperature when using room temperature fluid and that warmed irrigation fluid decreased perioperative hypothermia, especially in elderly patients.¹² The osmolality of irrigation fluid may also be an important factor to consider when performing arthroscopy. An in vivo animal study was performed to evaluate the safety of using a hyperosmolar irrigation fluid (600 mOsm/L) compared with normal saline (300 mOsm/L) prior to designing a study on humans. They tested bilateral shoulders in dogs and found the mean percentage change in shoulder girth to be higher in the isotonic control group (13.3%) compared with the hyperosmolar group (10.4%). There were no detrimental effects on chondrocyte viability with either solution.¹³ Future clinical studies on this topic may provide more information on ways to decrease fluid extravasation during prolonged arthroscopic procedures.

Instruments and Motorized Shavers

A variety of hand-operated instruments are used in arthroscopy. Many are generalized, but some instruments have been developed for particular joints or procedure-specific tasks. These instruments include various arthroscopic probes, graspers, baskets, scissors, knives, suture passers, and radiofrequency instruments. The actual instruments required for arthroscopy depend on surgeon preference, the joint involved, and the procedure being performed.

Arthroscopic Probe

The probe is the most important and used instrument in both diagnostic and therapeutic arthroscopy. Most probes are right angled at the working end, with a tip length of 4 mm that can be used to estimate intraarticular size, depth, and distance. Its primary purpose is palpation and identification of pathology, such as soft cartilage, loose bodies or cartilage flaps, meniscal tears, or insufficient tension of the anterior cruciate ligament. It is also a useful instrument for learning triangulation, planning surgical approaches, and positioning or retracting intraarticular structures or loose bodies. Care should be taken when palpating structures with the arthroscopic probe, because use of the tip may cause injury to tissues, and therefore use of the elbow for such activities is advisable.

Basket Forceps

Basket forceps are commonly used in procedures that involve the removal of intraarticular tissue, such as a meniscectomy. Basket forceps have a central opening and remove a portion of tissue that is then allowed to fall free into the joint, eliminating the need to remove the instrument from the joint to clear the basket each time (Fig. 8.3). These forceps typically range in size from 3 to 5 mm and are available in straight, left- or right-curved, and up- and down-angled varieties to facilitate work in different areas. Left- and right-curved varieties enable the surgeon to access

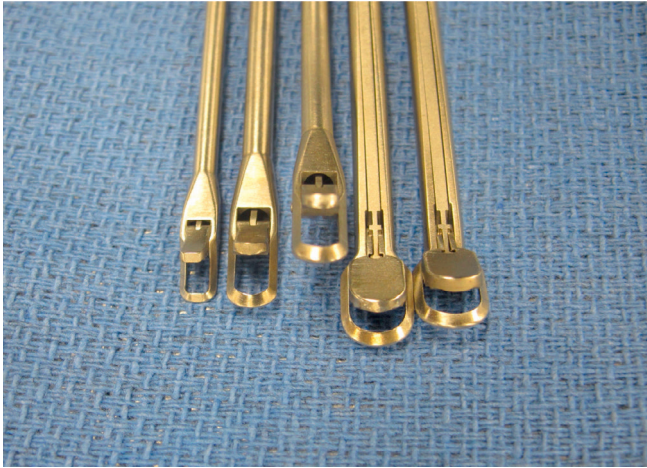


Fig. 8.3 Basket forceps for meniscal resection.

the anterior portion of the meniscus, whereas the up and down angled instruments permit trimming of the posterior meniscus around the femoral condyles. These instruments are also available in low-profile varieties for work in tight spaces, and some have hooked jaws that prevent tissue from slipping forward between the jaws when they are closed. Proper technique with these instruments is to use the jaws to remove small pieces of tissue because too large of a bite will strain the mechanical joints and may damage the instrument.

Grasping Forceps

Grasping forceps are used to securely hold, apply tension to, or retrieve intraarticular materials. These forceps are available in jaw configurations that include numerous serrated interdigitating teeth or with one or two sharp hooks at the distal end of the jaws. Configuration of the grasping jaws can also differ, in that varieties exist with both or only one mobile jaw. The graspers with forceps that have both jaws mobile are more suited to grasping and gripping larger intraarticular objects.

Scissors

Arthroscopic scissors are available in small and large sizes and with both straight and hooked blades to ensure an appropriate instrument for each situation. Hooked blades are easier to use compared with straight blades because they prevent the tissue from being pushed out from between the blades during the cutting motion. Curved and angled options, in which the shaft and blades are curved or the blades deviate from the shaft, respectively, are also available for difficult-to-reach resections.

Knives

Disposable arthroscopic knives are available in various shapes and sizes and should always be used with a cannula to protect the tissues during joint insertion and removal. Blade varieties include straight, curved, end cutting, and hooked, all of which allow the appropriate instrument for each situation. An important and useful characteristic of many arthroscopic knives is that the blades are magnetic to facilitate removal from the joint should it break.

Motorized Shavers

Motorized shavers consist of a pair of cylinders, one inside the other, each with a matching window. The inner cylinder spins within the outer and has an edge to both sides of its window that cuts tissue drawn into the outer cylinder by the attached suction. The removed tissue is then drawn down the inner cylinder, out of the joint, and into a suction trap. Commonly used motorized shavers range in diameter from 3 to 5.5 mm, but a 2-mm option is available for access to smaller or tight joints. Shavers with a larger diameter have larger windows and can expedite removal of tissue. Shavers also are available in various shapes and angles with different blades or burrs, all designed for specific purposes. Control of the motorized shaver is possible either by a hand or foot pedal and allows clockwise, counter-clockwise, and alternating rotation to improve the efficiency of tissue cutting and allow clearing of the blade. Practically, when the motorized shaver is in use, the blades should be in the field of view to prevent unwanted damage to normal tissues.

Cannulas and Switching Sticks

Cannulas and switching sticks of numerous sizes are available to assist with the insertion and use of instruments during arthroscopic procedures; they are all designed for use in specific joints. Cannulas are used to maintain the integrity of a portal into the joint so that the intervening soft tissues do not interfere with activities such as knot tying. They also allow easy insertion of instruments without trauma to the tissues through which the portal is made. Switching sticks are used to ease the passage of the arthroscope and cannulas into the joint in question by holding the path through the soft tissues into the joint so that a cannula or arthroscopic sheath can be inserted into the joint over it. When working in joints that are deep from the surface, switching sticks are particularly useful in maintaining these paths.

Electrocautery and Radiofrequency Instruments

Electrocautery and radiofrequency instruments can be used for cutting and shrinking soft tissues and can play an important role in maintaining hemostasis. A variety of these instruments have been developed for use in specific joints or for specific purposes, including some with flexible tips. Electrocautery uses heat directly from the probe to effect tissue changes and has been adapted for intraarticular use with various irrigation solutions.¹⁴

Radiofrequency instruments generate a high-frequency electromagnetic current at their tip, causing heat to be generated by the resistance to current passage through tissue.¹⁴ Monopolar and bipolar radiofrequency systems have been adapted for arthroscopic use. Monopolar systems allow current flow from the tip of the instrument to a grounding pad on the patient, using the path of least resistance, and generate heat at the tissue closest to the tip.¹⁴ Bipolar devices permit passage of current between two electrodes at the tip of the instrument, heating the tissue between them. The primary difference between the two radiofrequency systems is that monopolar instruments require a grounding pad to allow energy to pass from the body, whereas bipolar instruments pass energy between electrodes at the intraarticular site of interest.

The specific effect of radiofrequency instruments on tissues depends on the amount of heat transferred. Low levels of heat transfer cause collagen, which is the major constituent of most soft tissues, to denature and shrink by as much as 50% of its length.¹⁵ Higher levels of heat destroy collagen and are commonly used for tissue ablation and débridement.¹⁴ Use of these instruments intraarticularly may cause high fluid temperatures leading to cartilage damage or chondrolysis. Cadaveric studies have been performed measuring the effect of radiofrequency devices on intraarticular fluid temperature and cartilage. Increased duration of application, decreased distance between the probe and soft tissue, and decreased fluid rate showed temperatures deleterious to articular cartilage chondrocytes ($>50^{\circ}\text{C}$).^{16,17} Irrigation-fluid flow is critical for maintaining low fluid temperatures rather than the continuous or intermittent use of the thermal device, because decreased flow rates significantly increase the temperature of the irrigation fluid.¹⁸ The penetration depth and area affected are proportional to the power and temperature used.¹⁴ A level I study evaluated a standard radiofrequency device compared with a new plasma radiofrequency device and found no significant difference between the two systems with respect to temperature generation or operative time.¹⁹

ARTHROSCOPIC PRINCIPLES

Surgical Technique

“Arthroscopy is an art that requires three dimensional thought and vision.”

—Dr. Ronald Krinick, MD, New York, 2010

There are several skills required before one can successfully perform arthroscopy. The top four general skills are anatomic knowledge, tissue manipulation, spatial perception, and triangulation. The top two skills specific to arthroscopy are portal placement and triangulating the tip of the probe with a 30-degree scope.²⁰ Triangulation is the process by which your brain converts a two-dimensional arthroscopic image into a three-dimensional space, allowing your hands to manipulate an instrument entering from one portal in front of an arthroscopic camera with a specific degree of angulation from a separate portal. This is the most difficult aspect of arthroscopy initially and one that requires experience. Cadavers, arthroscopic computer simulators, and homemade arthroscopic simulators are great tools to have for acquiring these basic skills.²¹

Preoperative

It is important that the surgeon always meet the patient in the preoperative holding area prior to the patient being taken to the operating room. It is in this critical moment that the surgeon can have a clear discussion with the patient prior to any anxiolytic or narcotic medications on the correct procedure to be performed and to confirm the correct operative extremity. The operative extremity should be identified with confirmation by the patient and marked in the preoperative area with indelible ink, an ink that cannot be washed away, as recommended by the American Academy of Orthopaedic Surgeons.²² Marking the extremity with indelible ink prior to sterile prepping and draping has not been

shown to increase infection.²³ Knee arthroscopy is the most frequent “wrong side” surgery in malpractice claims.²⁴

A preoperative pause or “time out” should be enacted prior to the administration of anesthesia with the entire operative team present to confirm the correct patient, operative procedure, and operative site and to ensure that all required equipment is available. Discussing any concerns that members of the operative team may have regarding the case is also performed at that time.

Patient Positioning and Preparation

Patients undergoing arthroscopic surgery require careful positioning, especially during shoulder arthroscopy with the beach chair or lateral decubitus position (Fig. 8.4). Be cautious of the nonoperative extremities and ensure that they are secured. All bony prominences on the operative and nonoperative extremity should be well padded to avoid pressure on nerves that may result in a postoperative neuropraxia. If positioning the patient in the lateral decubitus position, an axillary roll should be placed three fingerbreadths distal to the axillary fold to avoid brachial plexus neuropraxia. Tourniquets are often used during knee arthroscopy. Zhang and Li performed a level one meta-analysis study to determine whether routine use of a tourniquet is beneficial in terms of arthroscopic visualization and operative time in knee arthroscopy. The study showed no significant difference in visualization or operative time between the tourniquet and nontourniquet group; however, multiple studies did suggest there were fewer visualization difficulties when a tourniquet was used during anterior cruciate ligament (ACL) surgery.²⁵ Tourniquets, if used, should be wide to distribute the compressive force over a larger surface area and should be limited to less than 90 minutes.²⁶ A tourniquet should be avoided if the patient has a history of thrombophlebitis, severe peripheral vascular disease, or endovascular stent placement which can be visualized on x-ray.

Surgical preparation and draping should be performed to ensure a sterile and sealed operative field. Surgical skin preparation is best performed with chlorhexidine and isopropyl alcohol because this combination has been shown to reduce bacterial



Fig. 8.4 Careful patient positioning and room setup for patient safety and clear visualization of the arthroscopic monitor.

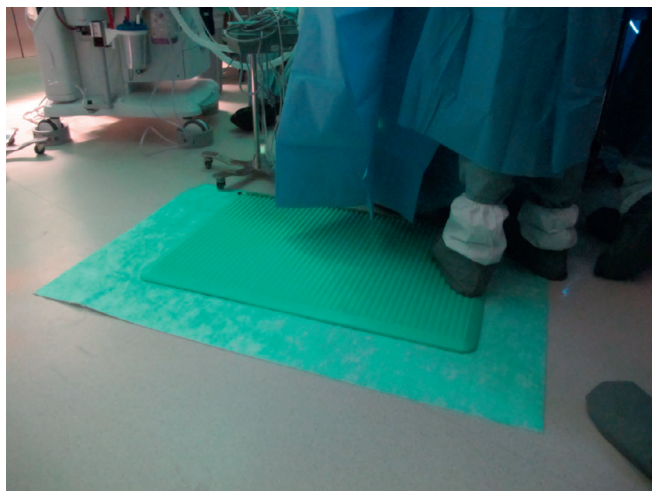


Fig. 8.5 The antislip irrigation collection mat prevents excess irrigation on the floor.

counts to a significantly greater degree than use of isopropyl alcohol, chlorhexidine, or povidone iodine alone.²⁷ Chlorhexidine has also been shown to have antimicrobial effects on the skin for up to 24 hours.²⁸ Benzoyl peroxide has been shown to be effective in the reduction of *Propionibacterium acnes*, lowering the risk of postoperative infection.²⁹ Sterile surgical draping technique will be determined by the joint undergoing arthroscopy and can be performed in many ways, according to the surgeon's preference. Prefabricated disposable draping is made specifically for a specific joint and has a fluid collection pouch to minimize fluid leakage onto the floor. There are also antislip mats with an integrated water collection system to assist with excess irrigation (Fig. 8.5).

Anesthesia

Local intraarticular anesthesia and injection of portal sites has been used alone for diagnostic arthroscopy,³⁰ although the use of a tourniquet or any bony procedures may not be tolerated in this setting. Recently, intraarticular Marcaine has also been shown to cause chondrolysis, making injection at the end of the procedure less favorable.³¹ Regional or spinal anesthetics can be used but carry the risk of hematoma and femoral neuritis. Femoral blocks or adductor canal blocks (saphenous nerve) may be used for regional anesthesia. General anesthesia is used most commonly and can be combined with regional techniques to enhance postoperative pain control. The patient, the procedure, and the preferences of the anesthesiologist and orthopedic surgeon should all be considered when deciding on the method of anesthesia (Fig. 8.6).

After anesthesia is induced, an examination of the joint in question should be performed in every case and compared with the contralateral side.

Arthroscopic Portals

Portal placement is vital for safe and effective arthroscopy. Knowledge of surface anatomy is key to successful placement of portals. The portals used will depend on the joint undergoing arthroscopy, and their placement also may be varied slightly by the procedure



Fig. 8.6 The line of the incision is injected with local anesthetic before making the incision.

being performed. For diagnostic arthroscopy, a minimum of an arthroscopic portal and a working portal (for instruments such as the probe) are required. Depending on the procedure, additional accessory portals may be needed. The arthroscope sheath generally is inserted first using a blunt trocar, which is then replaced by the arthroscope. The blunt trocar also may be used to define additional portals prior to their use for the insertion of instruments.

Diagnostic Arthroscopy

One must routinely perform a systematic diagnostic arthroscopy prior to any therapeutic intervention. This allows visualization of the entire joint, giving you a thorough picture of what is normal and what is not. A systematic diagnostic arthroscopy has been described for each joint. This procedure provides confirmation of the preoperative diagnosis and surgical plan, while allowing the opportunity for any required modification. It also minimizes operative time and the incidence of incorrect procedures or missed pathology.

ARTHROSCOPIC COMPLICATIONS

Arthroscopic complications are rare and are usually minor, but they do occur and surgeons and patients should be aware of them. The most critical complication to avoid is wrong side surgery. Preoperative planning and communication with the patient is paramount prior to the operation. Evaluating the patient in the preoperative holding area and marking the correct extremity with patient confirmation can eliminate this error.²² It is also helpful to have the radiographic images or magnetic resonance imaging available for review in the operating room, which also aids in confirming the correct extremity and planned procedure. Careful positioning of the patient for arthroscopy, surgical technique, and a comprehensive knowledge of the anatomy are important to decrease inadvertent damage to extra- and intra-articular structures. A preoperative “time out” should always be performed to verify the correct patient, site of surgery, procedure to be performed, and equipment required and to identify and address any concerns.

Arthroscopic complications can be joint specific. A large cross-sectional study with data taken from the ABOS database looked at the complications and complication rates of knee arthroscopies. They noted 4305 complications out of 92,565 knee arthroscopies (4.7%) from 2003 to 2009. The most common complication was infection, at 0.84%. The highest complication rates at 20.1% and 9.0% were for posterior cruciate ligament (PCL) and ACL reconstructions, respectively. Complications for meniscectomy, meniscal repair, and chondroplasty were 2.8%, 7.6%, and 3.6%, respectively.³²

In a review of elective shoulder arthroscopies, complications were noted in 93 of 9410 arthroscopies (0.99%). Of those 93 patients, 51 were considered a major morbidity (0.54%) and 42 considered a minor morbidity (0.44%). The most common complication was return to the operating room (29 cases, 0.31%). This was followed by superficial surgical site infection (15 cases, 0.16%), pulmonary embolism (6 cases, 0.06%), deep vein thrombosis (DVT) or thrombophlebitis (8 cases, 0.09%), peripheral nerve injury (1 case, 0.01%), and deep infections (1 case, 0.01%). The risk factors identified for complications included a history of chronic obstructive pulmonary disease (COPD), OR time greater than 1.5 hours, and an American Society of Anesthesiologists (ASA) class 3 or 4 compared with 1 or 2.³³

Harris et al. performed a systematic review of 92 studies and more than 6000 patients, collecting every reported hip arthroscopy in the literature prior to 2012. There were 512 complications out of 6134 patients (8.3%). The rate of major and minor complications was 0.58% and 7.5%, respectively. Minor complications were attributed to surgeon experience. The most common complications were iatrogenic chondrolabral injury (4.8%) during portal placement and temporary neuropraxia, at less than 1.4% (most commonly pudendal nerve and lateral femoral cutaneous nerve) secondary to prolonged or excess traction. The overall reoperation rate was 6.3% at a mean of 16 months postoperatively for conversion to total hip arthroplasty.³⁴

Intraoperative Complications

In general, the most common complication of arthroscopy is damage to intraarticular structures; however, this risk is lessened by the experience and care of the surgeon.³⁵ The cartilage surfaces of the joint undergoing arthroscopy are the intraarticular structures that are most at risk of complications, with damage being due to instruments or the arthroscope itself. Improper portal placement can also place intraarticular structures at risk during the insertion or removal of instruments or while working within the joint. If a poor portal is increasing the risk of complications, it is advisable to make a new properly placed portal. Working in tight joints or in areas of the joint that are difficult to reach will also place intraarticular structures at risk. The use of a different-angled scope, leverage, or traction to assist with opening the working area may decrease the risk of complications by ensuring appropriate room to manipulate the arthroscope and instruments. It is vital to maintain this working space until both the arthroscope and instruments are safely removed from the compartment in question to ensure that damage is avoided or minimized.

Extraarticular structures are also at risk during arthroscopy and can be a source of severe complications. Nerves and blood vessels are the primary extraarticular concern because they can be injured directly by portal placement or during arthroscopy as a result of their proximity to the work being performed. Similar to intraarticular complications, these complications also can be minimized with anatomic knowledge and appropriate exposure, being mindful of the location of extraarticular structures relative to the intraarticular work during arthroscopy, and using retractors for protection whenever possible. The use of the “nick and spread” method during portal placement will decrease the chance of injury. In this method, the skin alone is cut using a blade, and a hemostat is used to bluntly spread and create a path for insertion of arthroscopic instruments. Using blunt trocars to further define the portal prior to the insertion of instruments also may help protect neurovascular structures from injury. Appropriate articular distension and the avoidance of overly aggressive use of the motorized shaver near extraarticular structures will also minimize the chances of direct injury.

Indirect injury to extraarticular structures can occur as a result of pressure from the use of a tourniquet, fluid extravasation into surrounding tissues, or improper patient positioning. These structures also may be injured by traction that is required during arthroscopy of some joints. To avoid these complications, patients should be positioned with care and bony prominences should be padded, tourniquets should be applied properly and their use minimized (consider deflating them after 90 minutes), and traction should be used with care or avoided if possible. The use of gravity or low pump pressures may help minimize the extravasation of fluid into surrounding tissues, which can rarely cause a compartment syndrome if severe. Tourniquet paresis can be seen postoperatively but typically resolves spontaneously in days.

Instrument breakage can occur during arthroscopic procedures, particularly when instruments are older or used improperly. Small instruments are most likely to break, such as knife blades, or those with mobile parts, such as basket forceps. When an instrument breaks, intraarticular pieces must be removed. To remove the pieces, the outflow should be closed to stop turbulent flow and avoid movement of the broken piece, while maintaining distension of the joint. If the fragment moves and cannot be grasped before it becomes obscured in another part of the joint, radiographs may be useful for localization. For pieces that are difficult to grasp, suction or a magnet may be useful to remove or stabilize the piece so it can be grasped.

Postoperative Complications

The most common postoperative complication of arthroscopy is hemarthrosis,^{36,37} although it usually is of little consequence. Although certain arthroscopic procedures are known to be associated with a greater risk for this complication, a persistent or progressive hemarthrosis may require further vascular investigations or clotting studies to determine the cause.

A worrisome and potentially dangerous postoperative complication of arthroscopy is thrombophlebitis. Fortunately, this complication has been shown to be extremely rare in the upper extremity after arthroscopy, and it is uncommon in the lower

extremity.³⁶ Large retrospective studies have demonstrated that the incidence of venous thromboembolism (VTE) after knee arthroscopy is less than 1%.³⁸ Risk factors for DVT after arthroscopy remain controversial, but it is advisable to minimize both the tourniquet and operative times, mobilize patients postoperatively, and consider prophylaxis in patients who are known to be high risk or immobile.

Surgical infections are a rare but potentially devastating complication, and this is no different for arthroscopy. However, the rate of infection observed after arthroscopic surgery is a fraction of that seen in standard open procedures, likely because of many favorable patient and surgical factors. Arthroscopy typically is performed in young and healthy patients, using small incisions, with extensive irrigation for the majority of the short operative time. When postoperative infection is seen with arthroscopy, it is often in older patients who undergo complex procedures that require significant operative and tourniquet time, or it occurs as a result of breaks in sterile technique or instrument contamination.^{39–41} Most infections following arthroscopy are remedied with antibiotics and rarely require irrigation and débridement.

Preoperative prophylactic antibiotic use in arthroscopy remains controversial, and several studies in the last decade have shed light on the subject. Inappropriate use of antibiotics has led to an increase in antibiotic resistance and a rise in *Clostridium difficile* infections. Antibiotic stewardship programs (ASPs) are being implemented nationwide to increase the efficient use of antibiotics. The ASP at NYU Hospital for Joint Diseases recommend that perioperative antimicrobial prophylaxis is not required for arthroscopic surgeries without the use of an implant in patients with any risk factors.⁴² Multiple large studies have shown no association between the use of preoperative prophylactic antibiotics and postoperative deep or superficial infection rates.^{43,44} However, it may be advisable to consider antibiotic prophylaxis in patients at high risk for infection, such as those with diabetes, immune insufficiency, or skin problems.

Synovial herniation or fistula are both uncommon postoperative complications of arthroscopy. Synovial herniation is due to soft tissue or synovium protruding through the arthroscopic portal. These herniations typically do not require anything more than conservative measures, but they can be excised and closed if they are persistent and symptomatic. A synovial fistula is rarely seen but is thought to result from inadequate or lost closure because of a stitch reaction or abscess. Fistulas are treated with 7 to 10 days of immobilization and are unlikely to cause infection (although antibiotics should be considered) or require an additional procedure for excision and closure.

For a complete list of references, go to [ExpertConsult.com](https://www.expertconsult.com).

SELECTED READINGS

Citation:

Chierichini A, Frassanito L, Vergari A, et al. The effect of norepinephrine versus epinephrine in irrigation fluid on the incidence of hypotensive/bradycardic events during arthroscopic rotator cuff repair with interscalene block in the sitting position. *Arthroscopy*. 2015;31(5):800–806.

Level of Evidence:

I

Summary:

A double-blind, prospective, randomized controlled study of 120 patients comparing norepinephrine (0.66 mg/L) versus epinephrine (0.33 mg/L) use in shoulder arthroscopy irrigation fluid to assess the incidence of hypotensive/bradycardic events (HBEs) and efficacy in controlling intraoperative bleeding. Norepinephrine added to the irrigation fluid reduces the incidence of HBEs significantly compared with epinephrine and is as efficacious in controlling intraoperative bleeding thus maintaining clarity of the visual field.

Citation:

Hsiao M, Kusnezov N, Sieg RN, et al. Use of an irrigation pump system in arthroscopic procedures. *Orthopedics*. 2016;39(3):e474–e478.

Level of Evidence:

IV

Summary:

A review of arthroscopic irrigation systems and advancements in pump technology. The recent development of automated inflow and outflow mechanical irrigation pumps shows promising results with consistent flow, greater degree of joint distension, and improved visualization. Preliminary studies also show reduced operative times with automated systems secondary to improved visualization.

Citation:

Jackson RW. A history of arthroscopy. *Arthroscopy*. 2010;26(1):91–103.

Level of Evidence:

IV

Summary:

Professor Takaji from Tokyo is credited with being the first innovator and developer of arthroscopy. Other arthroscopy pioneers such as Dr. Bircher, Dr. Kreuzer, Dr. Burman, and Dr. Watanabe were integral in paving the road for arthroscopy. This paper gives a great historical recollection in the development of arthroscopy for the treatment orthopedic pathology.

Citation:

Kim YS, Lee JY, Yang SC, et al. Comparative study of the influence of room-temperature and warmed fluid irrigation on body temperature in arthroscopic shoulder surgery. *Arthroscopy*. 2009;25(1):24–29.

Level of Evidence:

I

Summary:

A prospective randomized study on 50 patients undergoing shoulder arthroscopy to determine the effect of room temperature versus warmed (37°C to 39°C) irrigation fluid on core body temperature. Hypothermia occurred more often in the room temperature fluid than the warmed fluid. Warmed irrigation fluid decreased perioperative hypothermia, especially in elderly patients.

Citation:

van Montfoort DO, van Kampen PM, Huijsmans PE. Epinephrine diluted saline—Irrigation fluid in arthroscopic shoulder surgery:

a significant improvement of clarity of visual field and shortening of total operation time. A randomized controlled trial. *Arthroscopy*. 2016;32(3):436–444.

Level of Evidence:

I

Summary:

A prospective, randomized, double-blind controlled trial evaluating the effect of epinephrine saline irrigation in shoulder arthroscopy procedures on clarity of view and operating time. Visual clarity was significantly better and total operating time was significantly shorter in the epinephrine group. No effect on the addition of epinephrine was seen on heart rate and blood pressure.

Citation:

Wyatt RWB, Maletis GB, Lyon LL, et al. Efficacy of prophylactic antibiotics in simple knee arthroscopy. *Arthroscopy*. 2017;33(1):157–162.

Level of Evidence:

IV

Summary:

A retrospective study of patients receiving simple knee arthroscopy between 2007 and 2012 to determine the association between the use of preoperative antibiotics and postoperative infection. Of the

40,810 simple arthroscopies, 80% received preoperative antibiotics and 20% did not. Deep infections accounted for 0.08% in the antibiotic group and 0.14% in the no-antibiotic group. Superficial infections accounted for 0.41% in the antibiotic group and 0.40% in the no-antibiotic group. There was no association between preoperative antibiotic use and postoperative deep or superficial infection rates.

Citation:

Zhang Y, Li L, Wang J, et al. Do patients benefit from tourniquet in arthroscopic surgeries of the knee? *Knee Surg Sports Traumatol Arthrosc*. 2013;21:1125–1130.

Level of Evidence:

I

Summary:

A meta-analysis of randomized controlled trials determining whether or not routine use of a tourniquet in knee arthroscopies is beneficial in terms of arthroscopic visualization and operative time. There was no significant difference in visualization or operative time between the tourniquet and nontourniquet group; however, multiple studies suggested that there was less visualization difficulties when a tourniquet was used with ACL surgery.

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